

Plant Station R17 - Assembly Manual

Controlled field	Value
Document number	PLANT-AM-001
Plant design revision	R17
Document issue	D02
Issue date	28 September 2026
Preceding document	PLANT-AM-001 R14 D01; original PLANT-AM-R14-UNO
Status	Issued for prototype review; physical acceptance open

Current mechanical configuration: R17 integrated body, R16 lid (R15 geometry), R13 fascia and planter. Electrical reference: frozen Uno UI v6 source and 27 September interface record. Fit, powered wet operation and unattended operation remain unaccepted.

R17 incorporates R15/R16 mechanical changes and the current Uno UI v6 interface. Previous issues are retained. Blank physical results remain open; no acceptance signature is inferred.

Revision control

Narrow parallel left-margin bars identify surviving changes against the verified matrix. Dates are documentary dates. R15/R16 source changes are incorporated without creating intervening manual issues.

Revision / class	Change and affected content
R5-R13 / source history	Earlier assembly and mechanical changes remain traceable through the preserved D01 matrix. Missing document editions are not invented.
R14 / preceding manual	AM R14 D01: Uno mounting and motor-disabled bench v2. TR's latest located same-family PDF is R13 D01; no TR R14 PDF is assumed.
R15 / 27 Sep 2026	Reinforced Uno bosses, thicker recessed-head lid and additional pump relief; source changes incorporated at R17, physical trials open.
R16 / 27 Sep 2026	Four relay bosses below Uno; source change incorporated at R17; actual underside and hole pitch unmeasured.
UI v6 / 27 Sep 2026	Three buttons and integrated lamps, enabled active-HIGH relay, 1-4 s request, soil display and 60 s backlight timeout. Firmware revision is independent of mechanical R17.
R17 / 28 Sep 2026	Mounted USB-C cut and pilots, adjacent diameter 10 DC extension hole; estimated dimensions authorised; coupon fit required.
D02 / 28 Sep 2026	Integrated current R17 document issue. Uses the established enhanced template, controls and cumulative revision bars. No R15/R16 manual issues created.

D02 / document control: new integrated R17 technical content in the established D01 A4 template. Pagination, new headings and editorial reflow alone receive no technical bar.

R5 is the earliest located assembly guide. R1-R4 and R9-R12 PDFs were not located in the previous inventory. R15/R16 are evidenced mechanical prototype changes, incorporated here without fictitious intermediate PDF/manual editions. Margin labels identify documentary changes, not physical test passes.

List of effective pages

All pages are effective at D02, dated 28 September 2026, for the R17 prototype. Section references provide navigation; technical change labels are controlled separately.

Page	Issue	Content
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11	D02	Section 7
12	D02	Section 8
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14	D02	Section 10
15	D02	Section 11
16	D02	Section 12
17	D02	Section 13
18	D02	Section 14
19	D02	Section 15
20	D02	Section 16

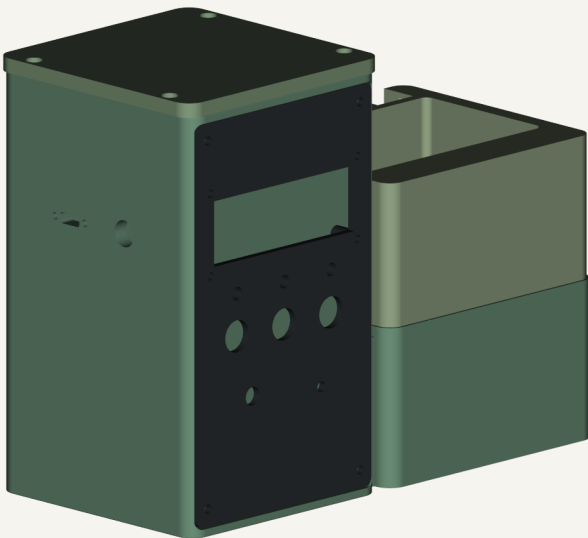
Section / page cross-reference

Section	Subject	Effective pages
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2	Printed parts and exact file selection	6
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01 Current status and configuration

R17 USB-C + DC PANEL assembly review



Provisional USB dimensions R16 relay / R15 mounts and lid retained

Actual R17 exported geometry with retained R16 lid and R13 fascia/planter. Components are omitted; this image does not establish fit.

R17

Use this integrated R17 manual for the current prototype build. R15/R16 changes are incorporated here. It replaces R14 instructions for the current assembly, while the R14 documents remain historical evidence.

CAUTION

Prototype status: digital geometry is checked; physical screw, connector, relay, pump-route and leak acceptance remain open. Do not infer those passes from a rendered assembly.

R17 | R15 | R13 | R10

Four main printed parts: R17 integrated body, R16 flush-head lid, R13 fascia and R13 planter. Select one knob variant. There is no separate battery tray or hose clip.

Terms: Confirmed = current file/check evidence; Reported = operator/source record; Assumed = explicit unmeasured geometry; Unknown = no adequate evidence; Not tested = no physical result. Firmware UI v6 is independent of design R17.

02 Printed parts and exact file selection

	Part / qty	File	Use / disposition
R17	M01 / 1	tower_sump_body_uno_ports_r17.stl	R17 body; 207 x 100 x 158 mm
	M02 / 1	fascia_charcoal.stl (R13)	Retained front panel, nominal 12 mm button holes
R15	M03 / 1	top_cover_flush_r16.stl	R15 lid geometry; 99 x 100 x 5.5 mm; assembled top z163.5
	M04 / 1	planter_sage.stl (R13)	Retained removable planter and two drains
	M05 / 1	One measured-shaft knob variant	Choose from R13 knob variants, not all eight
R17	T01 / first	power_ports_fit_coupon_r17.stl	6 thick x 60 x 24; print and fit USB and DC hardware
R15	T02 / first	boss_trial_pilot_2p8.stl (R15)	Full-depth screw trial; 3.0 alternative only after recorded failure
R15	T03 / first	lid_head_trial_r15.stl	Actual lid screw-head and seating trial
R15	T04 / first	pump_route_trial_r15.stl	Actual pump plus attached tubing clearance
R16	T05 / first	relay_mount_trial_r16.stl	Actual board, underside, screw and wire access trial

CAUTION

Do not print an older tower or lid alongside this set. The R14 carrier coupon is superseded for the revised boss/pilot acceptance; the R13 ESP32 carrier coupon is not an Uno coupon. Retained dimensions do not close actual fit.

03 Hardware, electronics and incoming evidence

	Item / qty	Specification basis	Evidence / action
	Uno / 1	Classic Uno R3, ATmega328P	Record actual clone/revision and four asymmetric holes
	Uno screws / 4	2.8 x 11 blind pilot; screw dimensions unknown	Trial fastener family/length; do not force a fit
R16	Relay / 1 + 4 screws	Nominal 50 x 26 PCB, 44.5 x 20.5 pitch, 3.1 holes	Photo/listing basis only; measure underside and holes
R15	Lid / 4 screws	3.4 through holes; 6.6 x 2.5 head pockets	Head/type/length unknown; use lid coupon
	Fascia / 4 screws	Retained printed pilots	Prior 3 x 8 thread-forming allocation is provisional
	LCD / 1 + 4 fixings	16 x 2 I2C; 75 x 31 centres	Actual backpack/pin markings; M2.5 stack provisional
R17	Buttons / 3	Four-wire momentary illuminated buttons	Identify contacts, LED polarity and current limit separately
	Potentiometer / 1	A1 dose control; shaft/nut unknown	Verify terminals and select actual shaft-fit knob
R17	USB-C / 1 + 2 screws	Image flange 20.1; PCB 9.1; depth 14.5	Cut/pitch estimated; panel electrical interface open
R17	DC extension / 1	Requested 10 mm panel cut	Thread, nut, grip range, polarity and cable unknown
	Pump / 1 + tubing	Separate supply reported 3.5 V	Rated voltage/current, envelope, outlet and tubing unverified
	Probe / 1	Installed 447 dry / 221 wet reference	Relative index; verify supply and installed probe
	Sounder / optional	10 mm case provision	Type/current unverified; no new wiring approved
	Holder / optional	Integrated insulated-holder saddle	Cell/charger/power path not accepted; leave cell out

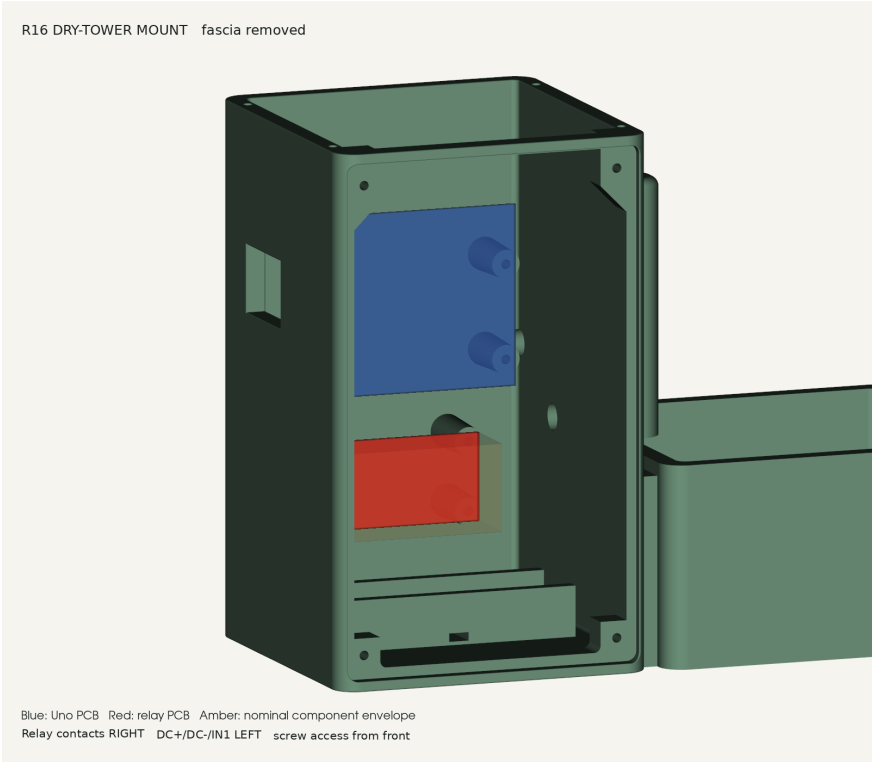
Use callipers, multimeter, hand drivers, small spanners and appropriate insulated harness tools. Printer, material, layer settings, fastener torque and instrument calibration status are Unknown until recorded. Do not use a powered driver for the printed pilot trials.

04 Fit trials and print hold points

- | | |
|------------|--|
| R17 | 1. Print the R17 port coupon. Put its broad 60 x 24 face on the bed, with 6 mm thickness and hole axes vertical. Inspect the slicer orientation and record the resulting bore dimensions. |
| R17 | 2. Fit USB-C and barrel hardware together: flange seats, mounting holes register, USB plug engages fully, DC nut/thread grip spans 6 mm, and both plug bodies and service leads clear. Stop on cracking, forcing or incomplete engagement. |
| R15 | 3. Trial the actual Uno screw in the R15 2.8 mm coupon. Inspect splitting, bottoming, retention, screw projection and board contact. The 3.0 mm trial is an alternative investigation; the R17 body remains 2.8 mm until explicitly regenerated. |
| R16
R15 | 4. Trial lid heads, pump/tube route and four-hole relay registration in their matching coupons. Record the actual screw, component, print material and photographs. These tests can run independently. |

R17	CAUTION Full-body print hold: resolve failed coupon interfaces, update the CAD parameter/source, rebuild and recheck exports. Slicing, physical manufacturing and acceptance are separate from the supplied digital checks.
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05 Uno and relay mounting; flush lid



R16 nominal PCB/component placement, retained inside R17. Proxies show approximate envelopes; this is not a fitted-hardware photograph.

- R14

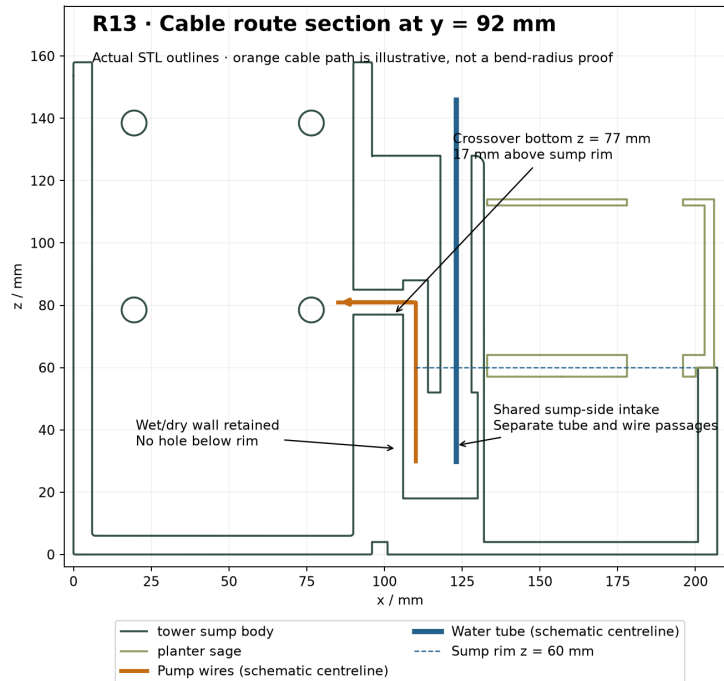
Uno hole centres x/z: (79.75,107.56), (79.75,79.62), (28.95,122.80), (27.68,74.54). Seat plane y76; board faces the removable fascia. Register all four holes without bending the PCB.
- R15

Revised Uno bosses: 8.5 mm tip / 10.0 mm root, 19 mm projection, 2.8 mm nominal blind pilot 11 mm deep from the seat. This addresses the reported R14 splitting but retention and print strength remain untested.
- R16

Relay PCB x21-71 / z33-59, seat y76; contacts to the right, logic terminals left. Hole centres x/z: (23.75,35.75), (23.75,56.25), (68.25,35.75), (68.25,56.25). Bosses match the Uno cone/pilot profile. Check underside solder and actual wires before tightening.
- R15

Lid: 5.5 mm thick with 3.4 mm through holes and 6.6 mm diameter x 2.5 mm deep pockets. Nominal head bearing plane remains z161; top z163.5. Trial head height and engagement, then hand-seat all four screws without distortion.

06 Wet-side pump, tube and sensor routes



Maximum fill level, cable jacket, connector size and minimum bend radius: UNKNOWN. No water-level approval.

Retained R13 route view: separate water tube and pump-wire passages. The lower pump relief is changed by R15; this historical route figure is not a complete R17 section.

- R6

With every supply disconnected, remove the planter and fit the actual pump with its tube attached. Reserved pump envelope 38.5 x 25.5 x 43 mm is a reference, not a measured supplied part.
- R12

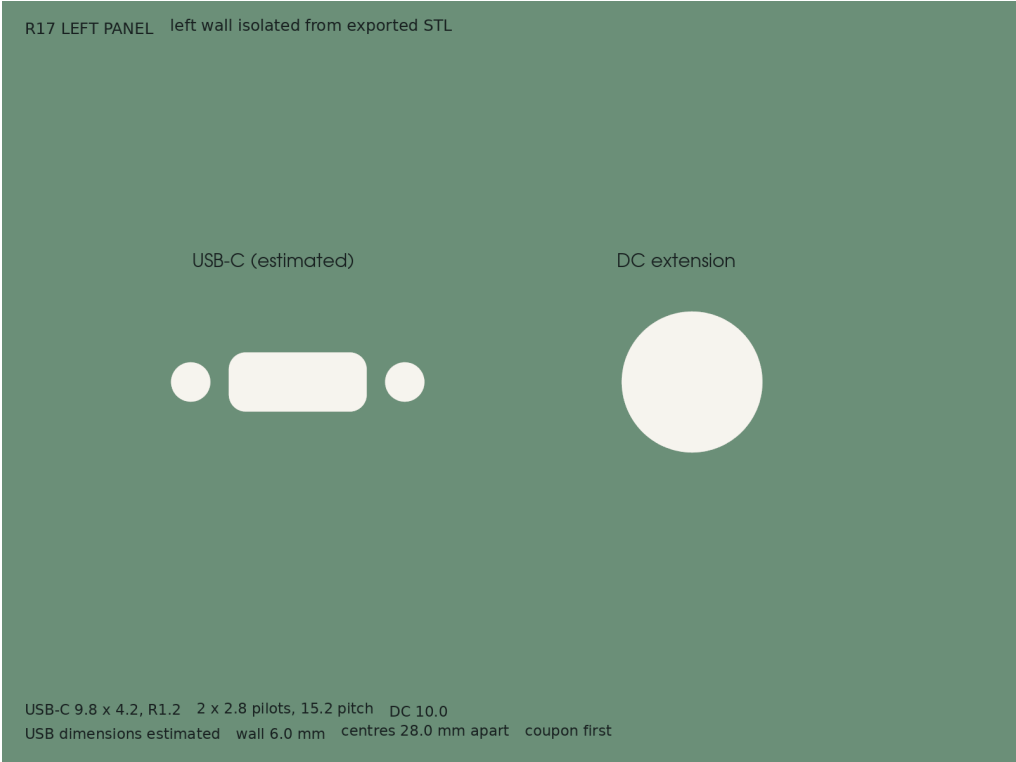
Feed tube through the 10 mm bore at x123/y92. Feed pump leads through the separate 8 mm bore at x110/y92 and its elevated crossover. Keep the water tube outside the electronics cavity; provide real-part strain relief and respect its bend radius.
- R15

R15 lower relief is x106.5-132, y83-94, z4-52. Sump floor stays 4 mm. Check the actual outlet/tube against the pump-route coupon and assembled body; preserve the dry-side partition.
- R13

Sensor entry is diameter 8, centre y65/z64, lower edge z60. Use the planter service relief, a removable service loop and suitable abrasion/drip protection. Lower edge z60 is a geometric datum, not a fill mark.
- R11

Tie an identified insulated holder into the integrated saddle only if used; slot 6 x 2.5, prior tie assumption 4.8 x 1.5. Keep the locking head accessible and all contacts clear. No cell or charging circuit is accepted by this mechanical provision.

07 Left USB-C and barrel-extension panel



Actual R17 left-wall mesh isolated at x <=6.01 to expose through-holes; other material is clipped for this view.

	Interface	Dimension / position	Status
R17	USB cut	9.8 x 4.2; corner radius 1.2; y63/z108.5	Authorised estimate
R17	USB pilots	2 x diameter 2.8 through; 15.2 pitch	Diameter matches PCB pilot convention; pitch estimated
R17	DC hole	Diameter 10.0; y35/z108.5	User-specified; no added allowance
R17	Wall / separation	6 thick; centres 28 apart along y	Selected placement; actual nut/plug envelopes unmeasured
R17	USB flange/PCB/depth	20.1 / 9.1 / 14.5	Image labels only; unmeasured actual part
R17	Mount the flange against the outside wall after the coupon passes. USB screws engage only the 6 mm wall, not an 11 mm blind boss; select length for the flange and check internal tips. Approximately 1.3 mm nominal material separates each pilot from the USB cut. Do not infer thread retention from the matching diameter.		
R17	CAUTION Electrical hold: identify connector pins, USB-C configuration-channel termination, data/power route and DC extension polarity/rating before soldering the installed harness. The image's 6-pin/5 A label is not verified. The cut does not authorise a 9 V connection to USB or a new direct 5 V feed to the Uno.		

08 Fascia components, harness and dry closure

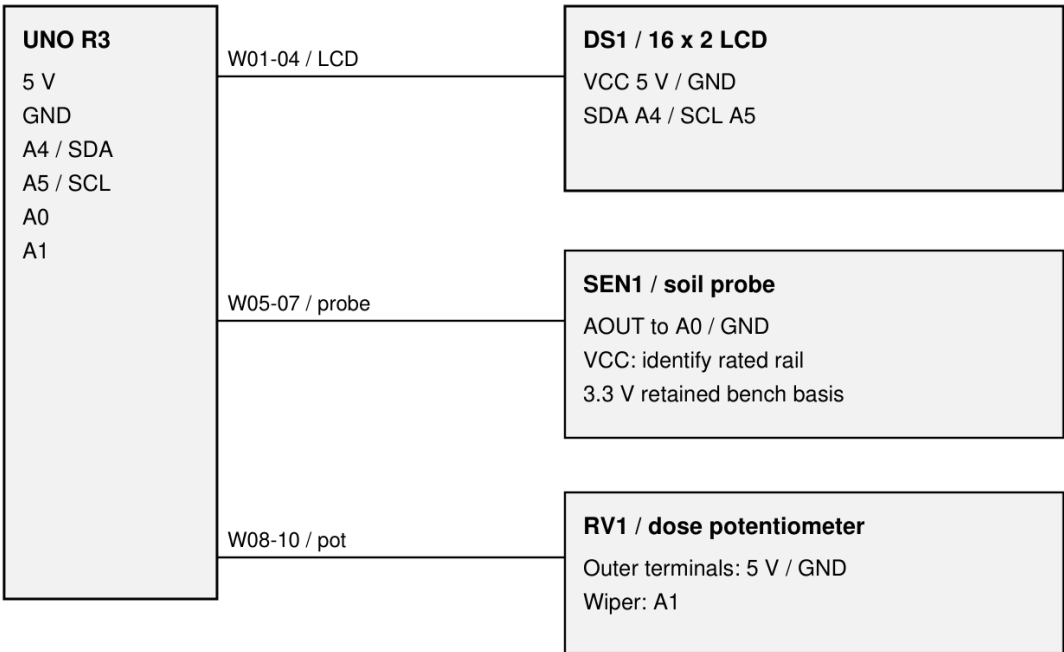


Retained R13 fascia and mounting pads. Printed geometry is retained; current control colours/functions are defined by E02, not historical artwork.

- | | |
|-----|---|
| R17 | <p>Fit the LCD to the 75 x 31 mm pattern behind the 71.4 x 24.6 aperture. Trial the actual fixings, head diameter and backpack stack; no PCB bowing or contact with screw tips.</p> |
| R8 | <p>Fit the green WATER, yellow NEXT/LAMP TEST and red STOP momentary buttons with their matching nuts. Verify nominal 12 mm holes against real hardware. Fit the potentiometer and select one shaft-fit knob; leave at least 0.5 mm axial clearance. No millilitre scale is valid until measured flow calibration.</p> |
| R8 | <p>The retained sound port/seal and LED collars are mechanical provisions. If used, apply compatible adhesive around case perimeters only; keep light surfaces, sound opening and conductors clear. Electrical type/rating remains unverified.</p> <p>Install labelled service loops and insulated joints clear of fasteners. Keep relay contact/pump leads separate from sensitive LCD/probe routes where practicable. Fit fascia, lid and planter by hand; confirm no trapping and full planter removal. No validated screw torque is supplied.</p> |

09 E01 - LCD, soil probe and dose potentiometer

E01 / LCD, probe and dose potentiometer



LCD VCC stays on regulated 5 V. D10 is unconnected. Verify actual module ratings.

Current logical route from the recorded Uno interface. Wiring is conditional on identified component ratings; shared return is a branch network, not a series chain.

R17 | LCD VCC to regulated Uno 5 V; GND to GND; SDA to A4; SCL to A5. D10 is unconnected. Backlight control uses the I2C backpack while LCD logic stays powered.

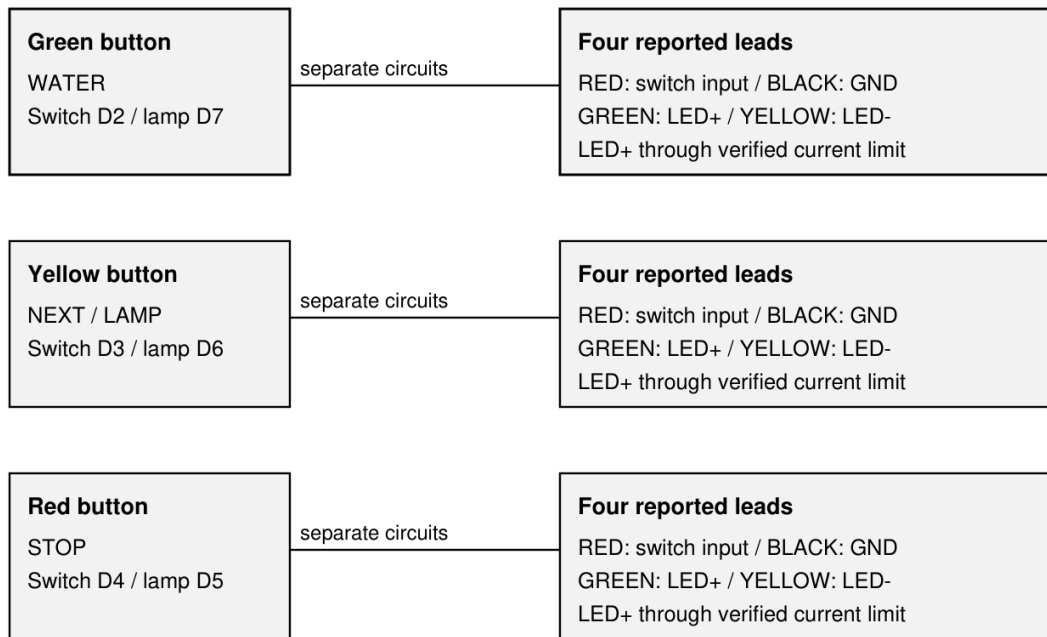
R17 | Soil AOUT to A0 and dose-pot wiper to A1. Retained probe bench basis is 3.3 V subject to the actual sensor rating; current supply continuity has not been re-measured. Pot outer terminals use 5 V and GND; verify direction and terminals before connection.

CAUTION

Keep pump supply OFF for all logic, upload and LCD checks. Repair intermittent LCD initialisation/write failures before treating the displayed pages as accepted.

10 E02 - buttons, LEDs and numbered signal schedule

E02 / current button and indicator interface



Button colour is its function colour. Lead colours are reported and require individual checks.

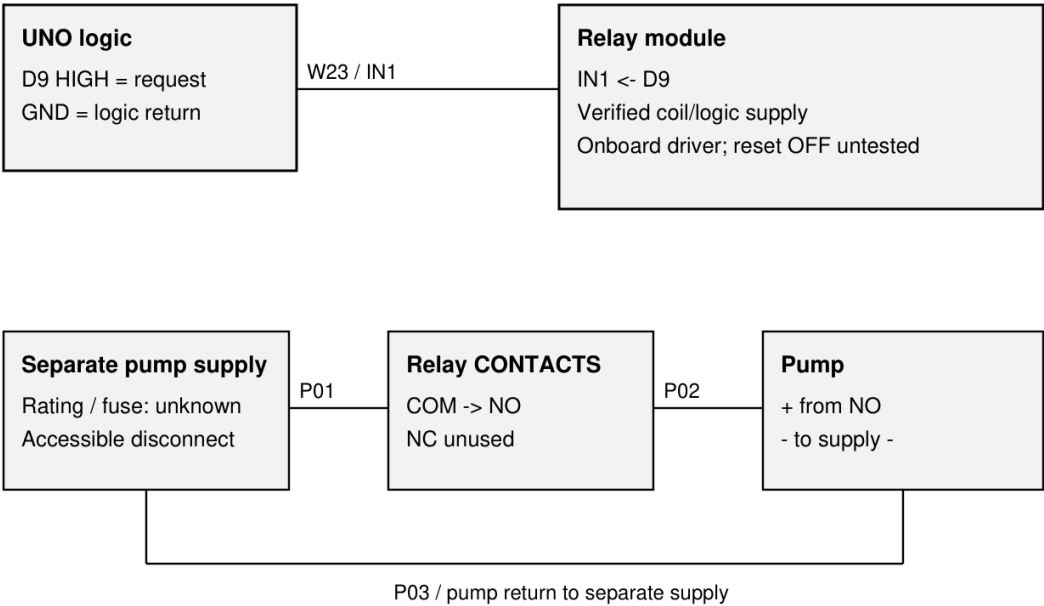
Function colour and reported lead colour are different attributes. Continuity/polarity must be confirmed for each physical button.

	Wires	From / to	Required identification
R17	W11 / W12	D2 / GND -> green WATER switch	Reported red / black leads; INPUT_PULLUP
R17	W13 / W14	D3 / GND -> yellow NEXT/LAMP switch	Reported red / black leads; INPUT_PULLUP
R17	W15 / W16	D4 / GND -> red STOP switch	Reported red / black leads; INPUT_PULLUP
R17	W17 / W18	D5 via limit / GND -> red lamp	Reported green LED+ / yellow LED-
R17	W19 / W20	D6 via limit / GND -> yellow lamp	Reported green LED+ / yellow LED-
R17	W24 / W25	D7 via limit / GND -> green lamp	Reported green LED+ / yellow LED-
R17	W23 / W26	D9 / logic GND -> relay IN1 / GND	Actual terminal labels and trigger current to verify

CAUTION
Each lamp needs its own verified current-limiting arrangement. The old R14 1 kOhm bare-LED example does not qualify the purchased illuminated buttons. Identify built-in limiting or select the external resistor from the actual LED evidence.

11 E03 - relay and complete supply schedule

E03 / relay logic and separate pump contact path



Contact path does not take pump power from Uno rails or a GPIO.
Pump voltage/current, protection and motor suppression remain to be identified and measured.

Logical relay and separate pump supply route. This is not a selected fuse, wire-size, contact-rating or suppression design.

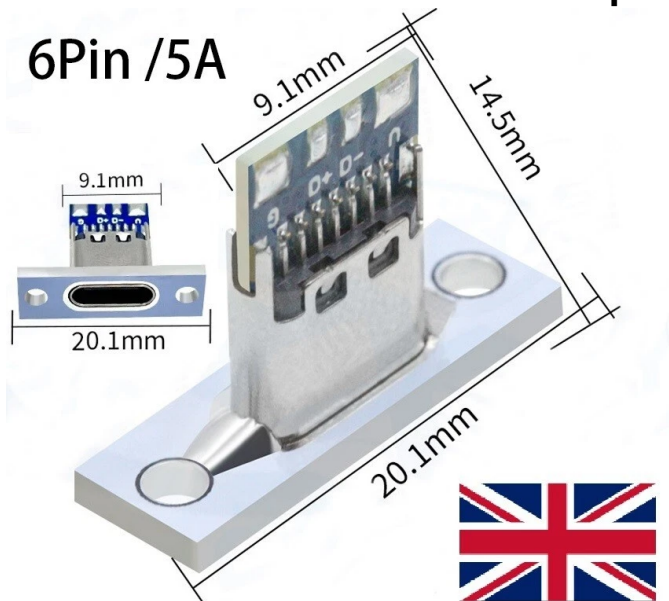
	Wire IDs	Connection	Disposition
R17	W01-04	5 V / GND / A4 / A5 -> LCD	As identified backpack markings
R17	W05-07	Rated probe rail / GND / AOUT-A0	Prior bench 3.3 V; confirm actual rating
R17	W08-10	5 V / GND / wiper-A1 -> pot	Verify outer terminals and direction
R17	W21 / W22	Uno 5 V / GND -> distribution	Record branch continuity and total logic load
R17	W27 / W28	Verified relay supply / GND -> module	Coil/input current and supply source unmeasured
R17	P01	Pump supply + -> fuse/disconnect -> COM	Supply/fuse/contact rating requires measurement
R17	P02 / P03	NO -> pump + / pump - -> supply -	Pump separate from Uno rails; NC unused

D9 is active HIGH in the current source; LOW at normal idle. During reset/bootloader it is high impedance. Measure relay OFF across reset and supply ordering, input current and contact state; provide a proven external default-OFF arrangement if necessary.

CAUTION
STOP cancels the software request. It cannot isolate a welded relay contact or a failed controller. Keep an accessible physical pump-power disconnect and rated protection. Identify the actual motor and suppression before a powered trial.

12 USB and barrel extension - open electrical interface

6Pin /5A



UK DISPATCH
FAST AND FREE P&P

Original user-supplied component image. Labelled geometry is reference evidence; its seller rating and pin functions have not been independently qualified.

R17 The R17 cuts support the intended alternative USB and Uno barrel-input connections. No USB-C solder pin mapping, cable adaptation or DC plug selection is approved by this document. Use the identified onboard USB connection for dry bench checks until A17-07 closes.

	Required evidence	Close A17-07 by recording
R17	USB component identity	Exact variant, connector pin map and measured shell/flange/fixing dimensions
R17	USB power/data route	Continuity, configuration-channel termination, intended data capability and supply compatibility
R17	Barrel extension	Actual mating sizes, polarity continuity, rated cable/current and 6 mm panel grip
R17	Supply arrangement	Selected voltage for each board input, regulation/load evidence and intended one-source procedure
R17	Acceptance	Measured idle/load rails, no unwanted source coupling, insulated strain relief and retained wiring diagram

Owner: Unassigned. Target: Not set. Do not assume the phrase '9 V battery connection' specifies the purchased extension, its polarity or an accepted supply.

13 Firmware identity and operator behaviour

R17 Frozen electrical reference: PlantUno UI v6 for arduino:avr:uno. The supplied source enables RELAY_ENABLED=true and ActiveHigh on D9, with 1,000-4,000 ms request limits, sampled from A1 at WATER start. This manual does not upload or change firmware.

	Control / state	Current source behaviour	Physical evidence
R17	Green WATER	One bounded request; knob sampled once; release does not cancel	Actual duration/volume not measured
R17	Yellow button	Idle pages Status/Raw/Session; hold lamp test; cancels dose	Full pages and lamp behaviour open
R17	Red STOP	Cancels; release required to re-arm; hold 2 s clears volatile counters	Software cancellation, not isolation
R17	Moisture	Below 25% red blink; at/above 25% green; 500 ms stability	Relative calibrated index, not volumetric moisture
R17	A0 raw <=20	Fault indication; green OFF; CHECK SOIL A0	Fault classification is not a pump interlock
R17	Pump request ends	Immediate soil Status return; sensor fault remains explicit	Post-dose physical screen observation open
R17	Backlight	60 s idle off; buttons or >=8 ADC knob counts wake; pump stop restarts timer	Serial OFF/ON recorded; visual and post-dose timer open

R17 The 27 September upload record reports COM8, 14,950 flash bytes and 870 global RAM bytes. Current frozen source hash matches that record. LCD status 0 and serial idle-OFF/activity-ON were recorded; previous LCD faults remain unresolved. These are historical records, not new tests performed for this manual.

CAUTION

Re-detect the board/port if uploading later; obtain confirmation that the separate pump supply is OFF. Keep the two header files with the sketch and use the correct board profile. No sensor state initiates automatic watering.

14 Dry commissioning and stop criteria

Use PLANT-TS-001 R17 D02 for actual results. All supplies remain disconnected during continuity, contact, screw and strain-relief inspections. Pump supply stays OFF for powered logic and display checks.

- R17 | Verify each switch pair open/released and closed/pressed, each lamp polarity/limiting, LCD wiring and probe/pot connections. Record actual parts and voltages. Stop on uncertain pin labels, current limits or supply polarity.
- R17 | With pump supply OFF, confirm serial identity, relay request OFF at idle, stable LCD status 0 and no write failures. Check all three switches, page navigation, lamp test, STOP/release, A1 endpoints, 25% indications and the rail-low fault using a safe identified test arrangement.
- R17 | Observe 60 s backlight idle-off and button/knob wake; separately observe full 60 s after completed/cancelled requests. Confirm the soil Status page returns immediately on request end. Serial acknowledgement alone does not accept the actual LCD.

CAUTION
Stop on an unexpected relay contact closure, LCD fault, overheating, loose wiring, damaged printed support or trapped service lead. Record the defect and keep affected power isolated until corrected.

15 Wet acceptance, operation and care

Electrical/pump acceptance, material qualification and leak/fill testing are prerequisites for supervised wet trials. Define the method, maximum fill candidate and duration before recording a pass. No accepted fill volume exists here.

R13

With electronics removed or independently protected, test the actual printed sump for leakage and inspect dry boundaries. Verify drain-back headspace, both planter drains, riser behaviour, tube retention and splash/tilt ingress. Sensor lower edge z60 and sump rim are not approved fill marks.

R17

After ratings/protection/default-OFF checks, measure separate-supply pump contact timing and delivery at 1.0 and 4.0 s settings with the installed lift and tube. Define allowable duration/volume error from the intended requirement; those tolerances are not yet set.

Before each accepted supervised use inspect water, tube, drains, visible leaks and supply leads. Isolate all power before moving, cleaning or removing the planter. Empty before moving and support the complete base; do not immerse the electronics tower.

Remove soil debris from both drains and pump inlet. Use cleaning materials and temperature limits compatible with the qualified print material. No unattended-operation approval or water-ingress rating is claimed.

16 Build record, open actions and source map

Build field	Actual recorded value
Build ID / operator / date	_____ / Unknown
Actual PCB / relay / pump identities	_____ / Unknown
Printer / material / slicer / saved job	_____ / Unknown
STL hash / source issue / orientation	_____ / Unknown
Fastener family / measured dimensions	_____ / Unknown
USB/DC component and wiring evidence	_____ / Unknown
Physical fit / leak / fill evidence	_____ / Unknown
Duration / volume results and tolerances	_____ / Unknown
Defects / deviations / action closure	_____ / Unknown
Reviewer / date / disposition	_____ / Unknown

R17

A17-01 through A17-13 are the shared acceptance IDs in PLANT-TS-001. Owner: Unassigned; targets: Not set. Blank fields are Unknown, never passes. Record evidence and explicit disposition before closing an action.

Sources: R13 retained CAD/readme; R14 Uno mounting evidence; R15 DESIGN_REVIEW and verification; R16 REVIEW and verification; R17 parameters, checks and attached image; current Uno sketch/headers; 27 September interface/upload record. The package contains frozen source copies, SHA-256 register and selected printable files.

Change bars use the verified control matrix. R17 electrical bars mean newly incorporated manual content from 27 September; they do not claim that UI v6 originated in the 28 September mechanical revision. R1-R4 and absent intermediate PDF editions remain undocumented.